



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)

Assignment-2

CSE 2216/CSI 218 (B) : Data Structure and Algorithms I Laboratory

Date : 1st July, 2024 (Summer 2024)

Doubly Linked List

Implement the following functions in a Doubly Linked List.

- a) Insert a node at the front
- b) Insert a node at the back
- c) Insert an node in the middle
- d) Delete the first node
- e) Delete the last node
- f) Delete a node from the middle
- g) Traverse the linked list

After implementing these functions try to get the following output for the given inputs

Input	Output
<i>print_list()</i>	<i>NULL</i>
<i>insert_at_front(1)</i> <i>print_list()</i>	1
<i>insert_at_back(2)</i> <i>print_list()</i>	1 2
<i>insert_in_middle(7,1)</i> <i>print_list()</i>	1 7 2
<i>insert_at_front(5)</i> <i>print_list()</i>	5 1 7 2
<i>insert_at_back(3)</i> <i>print_list()</i>	5 1 7 2 3
<i>delete_at_back()</i> <i>print_list()</i>	5 1 7 2

Input	Output
<i>delete_at_front()</i> <i>print_list()</i>	1 7 2
<i>delete(7)</i> <i>print_list()</i>	1 2

**You can take take from any online source or modify the code of singly link list provided in eLMS.

Practice Problems -

1. [Merge sorted Lists](#)
2. [Remove Duplicates](#)
3. [Reverse List](#)